Marmara University
Electrical and Electronics Engineering Department
Spring 2024
EE 3011 – Electronics I
Course Project – Due May 31, 2024

The main purpose of this simulation project is to study and verify the operation of an electronic circuit that has a specified function. The study includes simulation verification and hardware implementation.

Specifications

- Search the internet for reliable sites that give functional circuits with practical applications (example site below). Pick a circuit with a minimum of two discrete transistors (MOSFETs or BJTs). The circuit should not contain any ICs (only discrete components).
- Make sure that the circuit can be (1) simulated using LTspice and (2) easily implemented. Avoid complicated circuits with unusual components.
- Make sure that your circuit is unique (not used by other students in the class and not done in the lab experiments or discussed in class). There are plenty of choices for everyone.

Instructions

- Submit a report on your study and investigations of a circuit of your choice. The following tasks should be carried out.
  1) Use LTspice to verify the function of the circuit. The verification should cover different aspects of the circuit function, as follows.
     a) Circuit output (25%)
     b) Circuit limitations, range of operation and best operating conditions (25%)
     c) Possible modifications/improvements (20%).
  2) Test the hardware implementation of the circuit (20%).
  3) Discuss the work and give your opinion of the circuit based on investigations (10%).
- Maximum of three (3) students can work as a team.
- Prepare your project report (all in one file) in pdf format and upload to the site below.

Sample site: https://www.electronicshub.org/electronics-projects-ideas/

Upload link: https://www.dropbox.com/request/L8EGy3aJMBAhpuC6bPSv

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Show all work and analysis
Only pdf files are acceptable
Late submissions are checked but not graded
Copied submissions represent academic dishonesty and are disqualified